WORKSHEET 1.9

Unbounded Behaviour: Vertical and Horizontal Asymptotes

Unit 1: Limits and Continuity

Key ideas

- $x = c$ is a **vertical asymptote** if a one-sided limit at c is $\pm\infty$. Find the sign factor by factor.
- $y = L$ is a **horizontal asymptote** if $f(x) \to L$ as $x \to \infty$ or $x \to -\infty$.
- Rational functions at infinity: compare degrees; equal degrees give the ratio of leading coefficients.
- $\sqrt{x^2} = |x|$, which equals $-x$ when $x \to -\infty$. Growth: $\ln x \ll x^p \ll e^x$.

Worked examples

Example 1 · The sign of an infinite limit

Find $\lim_{x\to 1^-} \dfrac{x+3}{x-1}$ and $\lim_{x\to 1^+} \dfrac{x+3}{x-1}$.

Solution. The numerator is near 4. For $x < 1$ the denominator is a small negative number; for $x > 1$ a small positive number.

$-\infty$ from the left; ∞ from the right. The line $x = 1$ is a vertical asymptote.

Example 2 · Rational limits at infinity

Evaluate $\lim_{x\to\infty} \dfrac{2x^2-7}{3x^2+x}$ and $\lim_{x\to-\infty} \dfrac{x+1}{x^2+4}$.

Solution. First: equal degrees, ratio of leading coefficients. Second: the denominator has the higher degree.

$\dfrac{2}{3}$ and 0.

Example 3 · A square root as $x \to -\infty$

Evaluate $\lim_{x\to-\infty} \dfrac{\sqrt{x^2+9}}{2x-1}$.

Solution. $\sqrt{x^2+9} = |x|\sqrt{1+\frac{9}{x^2}} = -x\sqrt{1+\frac{9}{x^2}}$ for $x < 0$. The expression is $\dfrac{-\sqrt{1+\frac{9}{x^2}}}{2-\frac{1}{x}}$.

The limit is $-\dfrac{1}{2}$.

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Practice

CORE

1. Evaluate $\lim_{x\to\infty} \dfrac{5x^3 - x}{2x^3 + 4}$.

2. Evaluate $\lim_{x\to -2^+} \dfrac{1}{x+2}$.

3. Evaluate $\lim_{x\to\infty} \dfrac{4x+1}{x^2-3}$.

4. Evaluate $\lim_{x\to -\infty} e^x$.

EXAM LEVEL

5. Find both horizontal asymptotes of $f(x) = \dfrac{4e^x - 1}{2e^x + 3}$.

6. Find all vertical and horizontal asymptotes, and any holes, of $f(x) = \dfrac{x^2 - 4}{x^2 - 5x + 6}$.

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7. Evaluate $\lim_{x\to-\infty} \dfrac{3x+2}{\sqrt{4x^2+1}}$.

8. Evaluate $\lim_{x\to\infty} \left(\sqrt{x^2+8x}-x\right)$.

Question 9 · Free-response practice

Let $f(x) = \dfrac{3x^2-12}{x^2+x-6}$.

(a) Show that f has a removable discontinuity at $x=2$, and find $\lim_{x\to2} f(x)$.

(b) Write an equation for the vertical asymptote of the graph of f. Justify with a limit.

(c) Write an equation for the horizontal asymptote of the graph of f. Justify with a limit.

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Worked solutions

1. $\frac{5}{2}$.
2. ∞.
3. 0.
4. 0.
5. As $x \to \infty$: $\dfrac{4 - e^{-x}}{2 + 3e^{-x}} \to 2$. As $x \to -\infty$: $\dfrac{0 - 1}{0 + 3} = -\dfrac{1}{3}$. Asymptotes $y = 2$ and $y = -\dfrac{1}{3}$.
6. $f(x) = \dfrac{(x-2)(x+2)}{(x-2)(x-3)} = \dfrac{x+2}{x-3}$ for $x \neq 2$. Vertical asymptote $x = 3$; horizontal asymptote $y = 1$; hole at $(2, -4)$.
7. $\sqrt{4x^2 + 1} = -x\sqrt{4 + \frac{1}{x^2}}$ for $x < 0$, so the limit is $\dfrac{3}{-2} = -\dfrac{3}{2}$.
8. $\dfrac{8x}{\sqrt{x^2 + 8x} + x} = \dfrac{8}{\sqrt{1 + \frac{8}{x}} + 1} \to 4$.
9. (a) $f(x) = \dfrac{3(x-2)(x+2)}{(x+3)(x-2)} = \dfrac{3(x+2)}{x+3}$ for $x \neq 2$. So $\lim\limits_{x \to 2} f(x) = \dfrac{12}{5}$ exists while $f(2)$ is undefined: removable.

   (b) $x = -3$, because $\lim\limits_{x \to -3^+} f(x) = -\infty$ (numerator near -3, denominator small and positive).

   (c) $y = 3$, because $\lim\limits_{x \to \infty} f(x) = 3$.

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